

Department of Physics and Astronomy

First Year

2024-25

Tutorial Sheet 9

PHYS10302 Vibrations and Waves

PHYS10342 Electricity and Magnetism

PHYS10372 Mathematics 2

This sheet covers tutorials in weeks 10 of semester 2.

Problem solving can be a difficult skill to master. If you are finding it tough going, it may be helpful to try the following strategy:

1. Identify what the problem is about. Try to visualise the problem. Drawing a diagram may help.
2. Identify clearly what you are being asked to find.
3. What information have you been given? Sometimes even the simple task of writing down this information can help.
4. What (physics) principles and equations can you use?
5. How do you put points 1, 2, 3 and 4 together to get a solution? What maths will you need?
6. It is usually best to try and solve the problem algebraically, using symbols, even if you are given numerical values. This enables you to check at each stage and at the end, that you have correct and consistent dimensions. The last thing to do is to put numbers in, if a numerical answer is required.
7. Is the answer reasonable? If the answer is numerical, is the order-of-magnitude appropriate? Does your algebraic answer behave in the correct way when the variables take on very large or very small values, or in other limits?

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PHYS10302 Vibrations and waves

Week 10 questions

1. Light of wavelength 400 nm falls normally on a pair of slits, separated by 0.01 mm. Calculate the angles of the first 3 bright fringes. How many bright fringes are there altogether?
2. One way of figuring out the interference pattern for several slits is to sum a geometric series. Prove that the sum of the first N terms of a geometric series $a + ar + ar^2 \dots ar^{N-1}$ is $a \frac{r^N - 1}{r - 1}$.
3. Suppose we set up a Young's slits experiment with a ratio of slit separation to slit width of 4 to 1. The expected pattern of light and dark fringes is observed, but the 4th ($n = 4$) bright fringe is missing. Why?
4. In the lectures we showed the group velocity in a dispersive media is given by

$$v_g = \frac{\omega(k_2) - \omega(k_1)}{k_2 - k_1}.$$

By Taylor expanding $\omega(k_0 + \Delta k)$ and $\omega(k_0 - \Delta k)$, show that the group velocity can be written as

$$v_g = \left(\frac{d\omega}{dk} \right)_{k=k_0},$$

i.e. the group velocity is equal to the derivative of ω with respect to k , evaluated at the mean wavenumber $k_0 = (k_2 + k_1)/2$. Recall that $\Delta k = (k_2 - k_1)/2$.

5. This question is about water waves. The angular frequency ω of water waves of wavenumber $k = 2\pi/\lambda$ is given by

$$\omega^2 = gk + \frac{\gamma k^3}{\rho},$$

where g is the acceleration due to gravity and ρ and γ are the density and surface tension of water respectively.

- (a) Obtain an expression for the phase velocity of the waves.
- (b) Deduce expressions for the phase velocity of the waves in i) the limit of short wavelength ($\lambda \rightarrow 0$) and ii) long wavelengths ($\lambda \rightarrow \infty$).
- (c) Use your results from (b) to express ω as a function of k for both the long and short wavelength limits and hence obtain expressions for the group velocity of the waves in both limits.
- (d) Hence deduce the ratio of the group and phase velocities in i) the limit of short wavelengths and ii) the limit of long wavelengths.
- (e) For this question, use the long wavelength limit. Two boats, A and B, are stationary, floating in a bay. Boat A generates waves with a frequency of 2.3 Hz. What is their wavelength and phase velocity? Boat A is 40m away from boat B. It suddenly sinks. How long after the sinking do the waves from A stop arriving at B?

E&M Tutorial Sheet 9: Motional EMF

- 1) (Revision) Use Ampère's Law to calculate the magnetic field in a solenoid of length l , with N turns and carrying a current I .
- 2) A circular metallic disk of radius a rotates about an axis perpendicular to its centre with an angular velocity ω in a constant magnetic field, with strength B , which lies in the same direction as the rotation axis. This is known as Faraday's homopolar generator.
 - a) Explain why there is a potential difference between the edge of the disk and the centre.
 - b) Find size of the potential difference.
 - c) What might happen if this potential difference was used to drive an external circuit and this circuit was used to generate the uniform magnetic field? *The result is a so-called self-generating dipole. It turns out that the dimensions necessary for the self-sustaining generation are huge, but it is thought that this might be the mechanism that generates the Earth's magnetic field.*
 - d) What ultimately happens to the original kinetic energy of the disk? Assume that the external circuit has resistance R and that a current I flows through it (and therefore through the disk). Calculate the electrical power dissipated in the resistor, the torque acting on the disk due to the force on the current I and therefore the rate of change of the disk's kinetic energy.
- 3) The Earth's magnetic field is similar to that of a magnetic dipole as if it were inside the Earth orientated in a South-North direction¹. The field has the following characteristics in various places:
 - The field strength above the North Pole is 6×10^{-5} T.
 - At the equator it is half this value.
 - In London, the field strength is 5×10^{-5} T and the field lines make an angle of 60° to the horizontal.

An aeroplane has a wing span of 80m, a length of 60m and the fuselage has a height of 8m from top to bottom. Over which dimension is a motional EMF generated if it flies horizontally at 700 km hr^{-1} (200 m s^{-1}) in the following circumstances. Estimate the relevant potential difference in each case.

- a) Directly over the North Pole.
- b) Northwards over the Equator.
- c) Eastwards over the Equator.
- d) Northwards over London.

Hints: In each case first think about the direction of the magnetic field. Then in each case apply the Lorentz force law to work out in which direction charges are being pushed. The charges will move until an electric field builds up opposing their flow. Use the Lorentz Force Law to work out the size of this field. Make the approximation that the electric field is constant and hence $\Delta\phi = E \cdot d$, where d is the appropriate dimension of the plane.

¹ Note that the compass "North Pole" of the Earth is defined to be the direction towards which the "North" pole of a magnetic needle is attracted. Therefore, confusingly, the magnetic South Pole of the Earth is near to the geographic North Pole. You may find it useful to sketch the field lines of the Earth's magnetic field, with a dipole pattern and the field lines emerging near the South Pole and circulating around to re-enter near the North Pole.

PHYS10372 Mathematics 2: Tutorial-09 Questions

These questions cover the material from week-9 of PHYS10372: Hermitian, Unitary matrices, the matrix inverse.

1.

a. Demonstrate which of the following three matrices are Hermitian.

$$A = \begin{pmatrix} 2 & i & 0 \\ i & 3 & 4 \\ 0 & 4 & 1 \end{pmatrix}, \quad B = \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 3 & 5 & 6 \end{pmatrix}, \quad C = \begin{pmatrix} 3 & 2-i & 1 \\ 2+i & 5 & 3 \\ 1 & 3 & 4 \end{pmatrix}$$

b. Demonstrate which of the following three matrices are Unitary.

$$A = \begin{pmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} & 0 \\ -\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} & 0 \\ 0 & 0 & 1 \end{pmatrix}, \quad B = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 1 \\ 0 & -1 & 1 \end{pmatrix}, \quad C = \begin{pmatrix} \frac{1}{2} & \frac{1}{2} & \frac{1}{\sqrt{2}} \\ \frac{1}{2} & \frac{1}{2} & -\frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} & 0 \end{pmatrix}$$

c. Show that the matrix: $X = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & i & 0 \\ i & 1 & 0 \\ 0 & 0 & \sqrt{2} \end{pmatrix}$ is unitary and then that the operation: $X^\dagger A X$ produces a diagonal

matrix where $A = \begin{pmatrix} 0 & -i & 0 \\ i & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix}$.

2. The Hadamard gate (H) and the Pauli-X gate (X) in quantum computers are represented by the matrices:

$$H = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}, \quad X = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}.$$

A pure qubit state $|\psi\rangle$ is a coherent superposition of two basis states i.e. $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$ and $|\alpha|^2 + |\beta|^2 = 1$. For example the $|0\rangle$ qubit configuration (basis) could correspond to an electron spin being up and the $|1\rangle$ with it being down such that $|0\rangle$ corresponds to the vector $\begin{pmatrix} 1 \\ 0 \end{pmatrix}$ where the upper element is spin-up and the lower spin-down and thus that $|1\rangle$ corresponds to the vector $\begin{pmatrix} 0 \\ 1 \end{pmatrix}$. The electron's spin can exist in a superposition of the two states, allowing it and hence the associated qubit to represent more than just classical bits.

a. Demonstrate that H and X are unitary.

b. Determine what state is produced if we apply the Hadamard gate to $|0\rangle$ i.e. to $\begin{pmatrix} 1 \\ 0 \end{pmatrix}$. Express this in terms of $|0\rangle, |1\rangle$.

c. Determine the state produced if we then apply the Pauli-X gate to the state produced from the Hadamard gate.

d. Determine what state is produced if we apply these gates in the reverse order, X then H.

3. The EM fields matrices (tensors) in two different frames are written as:

$$F = \begin{pmatrix} 0 & E_x & E_y & E_z \\ -E_x & 0 & B_z & -B_y \\ -E_y & -B_z & 0 & B_x \\ -E_z & B_y & -B_x & 0 \end{pmatrix}, \quad F' = \begin{pmatrix} 0 & E'_x & E'_y & E'_z \\ -E'_x & 0 & B'_z & -B'_y \\ -E'_y & -B'_z & 0 & B'_x \\ -E'_z & B'_y & -B'_x & 0 \end{pmatrix}$$

and the Lorentz Transform for a boost in the x -direction with velocity, $\beta = \frac{v}{c}$ is: $L = \begin{pmatrix} \gamma & -\gamma\beta & 0 & 0 \\ -\gamma\beta & \gamma & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$

Determine the matrix (LF) and then multiply by L^T to determine the transform LFL^T . This transform gives F' . By equating the terms in LFL^T with those in F' , write down what the components of E and B in the primed frame are with respect to those in the un-primed frame.

4. Using the co-factor method determine the inverse of the following matrices.

$$A = \begin{pmatrix} 1 & 1 & -5 \\ 2 & 1 & -8 \\ -1 & -1 & 6 \end{pmatrix}, \quad B = \begin{pmatrix} 1+i & 2 \\ 3i & 1-i \end{pmatrix}, \quad C = \begin{pmatrix} 3 & 1 & 2 \\ 6 & 2 & 4 \\ 5 & -1 & 2 \end{pmatrix}$$

5. Cryptography relies heavily on matrices (linear-transforms) and also non-linear transforms to add confusion. We have an encrypted string "069099126037036044118083136" and a private encryption key. The private encryption key is a 3×3 matrix, K . Any encrypted string e is related to the original un-encrypted string o by the relation $e = Ko$. Our strings are defined as a sequence of 3×1 vectors where the elements are numbers, mod-26, i.e. "ALFRED" would be "0 11 5 17 4 3" and becomes the two vectors

$$\begin{pmatrix} 0 \\ 11 \\ 5 \end{pmatrix} \text{ and } \begin{pmatrix} 17 \\ 4 \\ 3 \end{pmatrix}.$$

Our non-linear transform, N , simply squares the value of a single number. In reality this non-linear transform would be more complex e.g. involving a private look-up-table and operations on adjacent numbers. Our encryption algorithm is thus a four stage process:

1. Convert un-encrypted string to a set of 3×1 integer vectors using modulo-26,
2. Encrypt each of these vectors via the encryption matrix (key) K ,
3. Perform a non-linear transform: square the middle element of the second encrypted vector,
4. Convert these vectors into a string: each element of the string is the element of the vector padded with zeros

The key matrix is $\begin{pmatrix} 2 & 1 & 3 \\ 4 & 0 & 1 \\ 5 & 2 & 1 \end{pmatrix}$, which you will need to invert.

All values in the encrypted vectors have values less than 1000 and thus the vector $\begin{pmatrix} 27 \\ 132 \\ 7 \end{pmatrix}$ would be written as "027132007".

The encrypted vectors are added to the string to the right i.e. the first vector forms the first (leftmost) elements of the string.

Determine what the original un-encrypted string is.